

BRAGA'S WEAK p -BANACH-SAKS PROBLEM FOR WEAKLY SEQUENTIALLY CONTINUOUS COARSE EMBEDDINGS: AUDIT, MODULUS TRANSFER, AND CURRENT STATUS

RESEARCH NOTE PREPARED FROM THE SUPPLIED SOLVER OUTPUT

ABSTRACT. Braga asked whether the weak p -Banach-Saks property passes from a Banach space Y to a Banach space X whenever X admits a weakly sequentially continuous coarse Lipschitz embedding into Y . We audit a supplied partial proof and find it correct, apart from a misnumbered citation and two minor omitted justifications. The proof gives permanence under the stronger hypothesis that Y has the tree- p -Banach-Saks property.

We place that argument in a more general modulus-transfer statement. If Y has tree- p -Banach-Saks constant C and $f : X \rightarrow Y$ is a weakly sequentially continuous coarse embedding, then, for every scale r with $\rho_f(r) > 0$, every normalized weakly null tree in X has a branch satisfying

$$\rho_f \left(r \left\| \sum_{j=1}^k x_{n_1, \dots, n_j} \right\| \right) \leq C \omega_f(r) k^{1/p}.$$

For coarse Lipschitz embeddings this yields tree- p -Banach-Saks permanence with an explicit sharp-at-infinity constant. We also give a direct positive answer under the original weak p -Banach-Saks target assumption when the embedding has controlled additive defect.

A literature search through June 21, 2026 found no published proof or counterexample to Braga's question. The original problem therefore appears to remain open. The central obstruction is genuine: weak p -Banach-Saks controls one weakly null sequence, whereas the nonlinear telescoping argument creates a weakly null tree, and Braga constructed reflexive weak p -Banach-Saks spaces with no tree- q -Banach-Saks property for any $q > 1$.

1. THE PROBLEM AND A RESULT LEDGER

For a map $f : X \rightarrow Y$ between Banach spaces, write

$$\begin{aligned} \omega_f(t) &= \sup \{ \|f(x) - f(y)\| : \|x - y\| \leq t \}, \\ \rho_f(t) &= \inf \{ \|f(x) - f(y)\| : \|x - y\| \geq t \}. \end{aligned}$$

The map is a *coarse embedding* if $\omega_f(t) < \infty$ for every t and $\rho_f(t) \rightarrow \infty$ as $t \rightarrow \infty$. It is a *coarse Lipschitz embedding* if there are $a, b > 0$ and $A \geq 0$ such that

$$(1) \quad a \|x - y\| - A \leq \|f(x) - f(y)\| \leq b \|x - y\| + A \quad (x, y \in X).$$

It is *weakly sequentially continuous* if $x_n \rightarrow x$ implies $f(x_n) \rightarrow f(x)$.

Braga's Problem 8.6 asks the following [4, Problem 8.6].

Question 1.1 (Braga). Let $1 < p < \infty$. Suppose that Y has the weak p -Banach-Saks property and that there is a weakly sequentially continuous coarse Lipschitz embedding $f : X \rightarrow Y$. Must X have the weak p -Banach-Saks property?

The conclusions established in this note are as follows.

- (i) The supplied partial proof is valid: tree- p -Banach-Saks in Y implies tree- p -Banach-Saks in X under such an embedding; see Corollary 4.2.

- (ii) The argument admits the scale-sensitive form in Theorem 4.1, valid for arbitrary coarse embeddings and for $1 < p \leq \infty$.
- (iii) Under the original weak p -Banach-Saks assumption on Y , a positive answer follows if the nonlinear embedding is sufficiently additive at suitable large scales; see Theorem 5.1. In particular it follows for embeddings with bounded Cauchy defect; see Corollary 5.2.
- (iv) No complete proof or counterexample to Question 1.1 was located in the literature. The tree-versus-sequence gap prevents the supplied argument from settling the original question.

2. BANACH-SAKS DEFINITIONS AND THE TREE GAP

Definition 2.1. Let $1 < p \leq \infty$ and use $k^{1/\infty} = 1$.

- (i) A Banach space Z has the *weak p -Banach-Saks property with constant C* if for every weakly null sequence $(z_n) \subset B_Z$ and every $k \in \mathbb{N}$ there is an infinite $M \subset \mathbb{N}$ such that

$$\left\| \sum_{j=1}^k z_{n_j} \right\| \leq Ck^{1/p} \quad \text{whenever } n_1 < \dots < n_k \text{ belong to } M.$$

- (ii) A family $(z_s)_{s \in [\mathbb{N}]^{\leq k}}$ is a *weakly null tree* if, for every node s of length less than k , the successor sequence $(z_{s \smallfrown n})_{n > \max s}$ is weakly null.
- (iii) The space Z has the *tree- p -Banach-Saks property with constant C* if every normalized weakly null tree $(z_s)_{s \in [\mathbb{N}]^{\leq k}} \subset S_Z$ has a branch $n_1 < \dots < n_k$ for which

$$\left\| \sum_{j=1}^k z_{(n_1, \dots, n_j)} \right\| \leq Ck^{1/p}.$$

The tree property implies the weak property by applying it to the tree whose value at $(n_1, \dots, n_j)$ is the n_j th term of a weakly null sequence [4, Remark 6.4]. The converse is false in a very strong form: for every $1 < p < \infty$, Braga constructed a separable reflexive space with the weak (indeed alternating) p -Banach-Saks property but without tree- q -Banach-Saks for any $1 < q < \infty$ [4, Theorem 7.1]. Later work places tree- p -Banach-Saks among a hierarchy of asymptotic and Hamming concentration properties and records strict inclusions even among reflexive spaces [6, 5].

We use the signed form of the tree property.

Proposition 2.2 (Braga's signed stabilization). *If Z has tree- p -Banach-Saks constant C , then for every normalized weakly null tree $(z_s)_{s \in [\mathbb{N}]^{\leq k}}$ there is an infinite $M \subset \mathbb{N}$ such that*

$$\left\| \sum_{j=1}^k \varepsilon_j z_{(n_1, \dots, n_j)} \right\| \leq Ck^{1/p}$$

for every $(n_1, \dots, n_k) \in [M]^k$ and every choice $\varepsilon_j \in \{-1, 1\}$.

This is Proposition 6.3 of [4], not Proposition 6.5 as stated in the supplied packet. It follows by applying the tree property to the finitely many levelwise sign changes and then using Ramsey stabilization.

Lemma 2.3 (Convexity of the sign cube). *Let $e_1, \dots, e_k$ belong to a normed space and let $t_1, \dots, t_k \in [-1, 1]$. Then*

$$\sum_{j=1}^k t_j e_j \in \text{conv} \left\{ \sum_{j=1}^k \varepsilon_j e_j : \varepsilon_j \in \{-1, 1\} \right\}.$$

Consequently, if all signed sums on the right have norm at most K , then $\left\| \sum_j t_j e_j \right\| \leq K$.

Proof. Choose independent signs ε_j with $\mathbb{E}\varepsilon_j = t_j$. Then $\sum_j t_j e_j = \mathbb{E} \sum_j \varepsilon_j e_j$, a convex combination of the sign vectors. $\square$

3. AUDIT OF THE SUPPLIED PARTIAL PROOF

The supplied argument has the right architecture. Given a normalized weakly null tree (x_s) in X , set

$$u_s = \sum_{t \preceq s} x_t, \quad y_s = f(u_s) - f(u_{s-}).$$

For a fixed predecessor s , one has $u_s + x_{s \smallfrown n} \rightharpoonup u_s$, so weak sequential continuity makes $(y_{s \smallfrown n})_n$ weakly null. Coarse Lipschitz control makes the increments semi-normalized after a harmless change of scale. Normalize the increments, use Proposition 2.2, return to the unequal increment lengths by Lemma 2.3, and telescope.

There are only three corrections or additions to make.

- (1) The cited signed result is Braga's Proposition 6.3.
- (2) If $y_{s \smallfrown n} \rightharpoonup 0$ and $\inf_n \|y_{s \smallfrown n}\| > 0$, then $y_{s \smallfrown n} / \|y_{s \smallfrown n}\| \rightharpoonup 0$: for every $y^* \in Y^*$, the numerator tends to zero while the denominator stays bounded away from zero.
- (3) The rescaled map $x \mapsto m^{-1}(f(mx) - f(0))$ is still weakly sequentially continuous because scalar multiplication and translation preserve weak convergence.

No other gap occurs. The next theorem gives a cleaner proof and retains the full compression and expansion moduli.

4. A GENERAL MODULUS-TRANSFER THEOREM

Theorem 4.1 (Modulus transfer). *Let $1 < p \leq \infty$. Suppose that Y has tree- p -Banach-Saks constant C and that $f : X \rightarrow Y$ is a weakly sequentially continuous coarse embedding. Fix $r > 0$ such that $\rho_f(r) > 0$. Then every normalized weakly null tree $(x_s)_{s \in [\mathbb{N}]^{\leq k}}$ in X has a branch $n_1 < \dots < n_k$ satisfying*

$$(2) \quad \rho_f \left(r \left\| \sum_{j=1}^k x_{(n_1, \dots, n_j)} \right\| \right) \leq C \omega_f(r) k^{1/p}.$$

The convention is $k^{1/\infty} = 1$.

Proof. Translation does not change ρ_f or ω_f , so assume $f(0) = 0$. For a nonempty node s , let

$$u_s = \sum_{t \preceq s} x_t, \quad u_\emptyset = 0, \quad y_s = f(ru_s) - f(ru_{s-}).$$

For every fixed predecessor s , the sequence $u_s + x_{s \smallfrown n}$ converges weakly to u_s . Hence $(y_{s \smallfrown n})_n$ is weakly null. Since adjacent partial sums are exactly distance r apart after scaling,

$$(3) \quad 0 < \rho_f(r) \leq \|y_s\| \leq \omega_f(r) < \infty.$$

It follows from the lower bound in (3) that $e_s = y_s / \|y_s\|$ is a normalized weakly null tree.

Apply Proposition 2.2 to (e_s) . On the resulting branch put

$$t_j = \frac{\|y_{(n_1, \dots, n_j)}\|}{\omega_f(r)} \in [0, 1].$$

Lemma 2.3 gives

$$\begin{aligned} \left\| \sum_{j=1}^k y_{(n_1, \dots, n_j)} \right\| &= \omega_f(r) \left\| \sum_{j=1}^k t_j e_{(n_1, \dots, n_j)} \right\| \\ &\leq C \omega_f(r) k^{1/p}. \end{aligned}$$

The sum on the left telescopes to

$$f\left(r \sum_{j=1}^k x_{(n_1, \dots, n_j)}\right) - f(0).$$

The definition of the compression modulus now yields (2). $\square$

The theorem can be read as an inverse-compression estimate. With the generalized inverse

$$\rho_f^{-1,+}(s) = \sup \{t \geq 0 : \rho_f(t) \leq s\},$$

Equation (2) gives

$$(4) \quad \left\| \sum_{j=1}^k x_{(n_1, \dots, n_j)} \right\| \leq \frac{1}{r} \rho_f^{-1,+}(C \omega_f(r) k^{1/p}).$$

This is meaningful without any affine hypothesis.

Corollary 4.2 (Coarse Lipschitz permanence). *Let $1 < p < \infty$. Assume that Y has tree- p -Banach-Saks constant C , and let $f : X \rightarrow Y$ be weakly sequentially continuous and satisfy (1). Then X has the tree- p -Banach-Saks property. More precisely, for every $r > A/a$, every normalized weakly null tree in X has a branch with*

$$(5) \quad \left\| \sum_{j=1}^k x_{(n_1, \dots, n_j)} \right\| \leq \frac{C(br + A) + A}{ar} k^{1/p}.$$

Consequently, for every $\eta > 0$, X has tree- p -Banach-Saks constant at most

$$\frac{bC}{a} + \eta.$$

In particular, X has the weak p -Banach-Saks property.

Proof. The affine bounds give $\omega_f(r) \leq br + A$ and, directly on the endpoint of the selected branch,

$$ar \left\| \sum_j x_{(n_1, \dots, n_j)} \right\| - A \leq \left\| f\left(r \sum_j x_{(n_1, \dots, n_j)}\right) - f(0) \right\|.$$

Combine this with the proof of Theorem 4.1 and use $k^{1/p} \geq 1$. Letting $r \rightarrow \infty$ gives the limiting constant. Tree- p -Banach-Saks implies weak p -Banach-Saks. $\square$

In the symmetric notation used in the supplied packet,

$$L^{-1} \|x - y\| - A \leq \|f(x) - f(y)\| \leq L \|x - y\| + A,$$

Corollary 4.2 gives constants arbitrarily close to $L^2 C$.

Corollary 4.3 (Power compression). *Under the hypotheses of Theorem 4.1, suppose additionally that*

$$\rho_f(t) \geq ct^\alpha - d \quad (t \geq 0)$$

for some $c > 0$, $d \geq 0$, and $\alpha > 0$. Then, for every r with $\rho_f(r) > 0$, the selected branch satisfies

$$\left\| \sum_{j=1}^k x_{(n_1, \dots, n_j)} \right\| \leq \frac{(C\omega_f(r) + d)^{1/\alpha}}{rc^{1/\alpha}} k^{1/(\alpha p)}.$$

Thus, if $\alpha p > 1$, the space X has tree- (αp) -Banach-Saks.

Proof. Insert the power lower bound in (2) and use $C\omega_f(r)k^{1/p} + d \leq (C\omega_f(r) + d)k^{1/p}$. $\square$

Remark 4.4 (The endpoint case). For $p = \infty$, Theorem 4.1 gives

$$\rho_f \left(r \left\| \sum_j x_{(n_1, \dots, n_j)} \right\| \right) \leq C\omega_f(r).$$

Since $\rho_f(t) \rightarrow \infty$, the branch sums are uniformly bounded independently of the height. Hence tree- ∞ -Banach-Saks passes under weakly sequentially continuous coarse embeddings. This is the tree formulation of the endpoint phenomenon in Braga's Theorem 6.11; tree- ∞ and upper ℓ_∞ tree estimates coincide in this setting [4, 6].

5. A DIRECT WEAK p THEOREM FROM ADDITIVE DEFECT

The tree proof cannot use weak p -Banach-Saks directly because its increments have different predecessors. There is, however, a complementary route: apply weak p -Banach-Saks to the single sequence $f(rx_n)$ and quantify the failure of f to commute with finite sums.

After translating f , put $\tilde{f} = f - f(0)$. For $k \in \mathbb{N}$ and $r > 0$, define the k -term additive defect on the unit ball by

$$(6) \quad \text{Def}_f(k, r) = \sup_{x_1, \dots, x_k \in B_X} \left\| \tilde{f} \left(r \sum_{j=1}^k x_j \right) - \sum_{j=1}^k \tilde{f}(rx_j) \right\|.$$

Theorem 5.1 (A defect criterion). *Let $1 < p < \infty$. Suppose that Y has weak p -Banach-Saks constant C , and let $f : X \rightarrow Y$ be weakly sequentially continuous and satisfy (1). Assume that there are $D \geq 0$ and scales $r_k \geq 1$ such that*

$$(7) \quad \frac{\text{Def}_f(k, r_k)}{r_k} \leq Dk^{1/p} \quad (k \in \mathbb{N}).$$

Then X has the weak p -Banach-Saks property, with constant at most

$$(8) \quad \frac{C(b + A) + D + A}{a}.$$

Proof. Let $(x_n) \subset B_X$ be weakly null and fix k . Define

$$g_k(x) = \frac{\tilde{f}(r_k x)}{r_k}.$$

Then g_k is weakly sequentially continuous, $g_k(0) = 0$, and

$$\|g_k(x_n)\| \leq b + \frac{A}{r_k} \leq b + A.$$

The sequence $(g_k(x_n))$ is weakly null in Y . The weak p -Banach-Saks property of Y supplies an infinite $M \subset \mathbb{N}$ such that

$$(9) \quad \left\| \sum_{j=1}^k g_k(x_{n_j}) \right\| \leq C(b + A)k^{1/p} \quad (n_1 < \dots < n_k \text{ in } M).$$

For such indices, (7) and (9) imply

$$\left\| g_k \left(\sum_{j=1}^k x_{n_j} \right) \right\| \leq (C(b + A) + D)k^{1/p}.$$

The lower affine estimate for g_k is

$$a \|x - y\| - \frac{A}{r_k} \leq \|g_k(x) - g_k(y)\|.$$

Taking $x = \sum_j x_{n_j}$ and $y = 0$, and using $A/r_k \leq Ak^{1/p}$, gives (8). $\square$

A simple concrete hypothesis implies the defect condition.

Corollary 5.2 (Bounded Cauchy defect). *Under the hypotheses on p, Y , and f in Theorem 5.1, assume*

$$(10) \quad D_0 := \sup_{u, v \in X} \|f(u + v) - f(u) - f(v) + f(0)\| < \infty.$$

Then X has the weak p -Banach-Saks property.

Proof. An induction gives $\text{Def}_f(k, r) \leq (k - 1)D_0$ for every k and r . Taking $r_k = k$ yields

$$\frac{\text{Def}_f(k, r_k)}{r_k} \leq D_0 \leq D_0 k^{1/p},$$

so Theorem 5.1 applies. $\square$

A genuinely asymptotic version is also enough. Define the local binary defect

$$\delta_f(R) = \sup_{\|u\|, \|v\| \leq R} \|f(u + v) - f(u) - f(v) + f(0)\|.$$

Corollary 5.3 (Sublinear additive defect). *Under the hypotheses on p, Y , and f in Theorem 5.1, suppose*

$$\lim_{R \rightarrow \infty} \frac{\delta_f(R)}{R} = 0.$$

Then X has the weak p -Banach-Saks property.

Proof. For $x_1, \dots, x_k \in B_X$, iterative addition at scale r only evaluates the binary defect on vectors of norm at most kr . Hence

$$\text{Def}_f(k, r) \leq (k - 1)\delta_f(kr).$$

For each $k \geq 2$, choose $R_k \geq k$ so large that

$$\frac{\delta_f(R_k)}{R_k} \leq \frac{k^{1/p}}{k(k - 1)},$$

and put $r_k = R_k/k \geq 1$. Then

$$\frac{\text{Def}_f(k, r_k)}{r_k} \leq k(k - 1) \frac{\delta_f(R_k)}{R_k} \leq k^{1/p}.$$

Take $r_1 = 1$ and apply Theorem 5.1. $\square$

Remark 5.4. Condition (10) says that the translated map $\tilde{f}$ is a quasi-homomorphism of the additive Banach groups; by classical Hyers–Ulam stability, such maps are close to additive maps under mild regularity. Theorem 5.1 and Corollary 5.3 are more flexible: they only ask for finite-sum additivity after division by scales that may depend on k . Thus they isolate a precise analytic quantity whose control would settle Question 1.1 for a given embedding.

6. WHY THE ORIGINAL ARGUMENT STOPS AT TREES

Let (x_s) be a weakly null tree in X and define increments

$$(11) \quad y_s = f\left(r \sum_{t \preceq s} x_t\right) - f\left(r \sum_{t \prec s} x_t\right).$$

Weak sequential continuity says that the successors of each fixed node form a weakly null sequence. It does *not* say that an enumeration of all y_s is weakly null: the predecessor changes while one enumerates the tree. Weak p -Banach-Saks can be applied separately at each node, but those local subsequences need not produce one branch on which the sum of increments is small.

Several standard-looking reductions therefore fail at exactly the same point.

- Enumerating tree nodes loses weak nullity because the base point of the increment moves.
- Applying weak p -Banach-Saks to the sequence $f(rx_n) - f(0)$ controls $\sum_j f(rx_{n_j})$, whereas coarse lower bounds concern $f(r \sum_j x_{n_j})$; their difference is the additive defect in (6).
- Hamming-cube or Hamming-graph averaging requires a concentration property stronger than weak p -Banach-Saks. Tree- p -Banach-Saks yields such concentration in reflexive or quasi-reflexive settings, but weak p alone does not currently do so [6].
- Passing to ultrapowers turns the levels of a weakly null tree into an asymptotic sequence, but weak p -Banach-Saks is not known to have the required ultrapower stability; tree properties are designed precisely to retain this asymptotic information.

The bad-tree example in Braga's Theorem 7.1 shows that this is not a cosmetic distinction. Any complete positive proof must exploit the additional "gradient" or cocycle relations enjoyed by (11), rather than merely treating it as an arbitrary weakly null tree. Conversely, a counterexample should combine a weak p -Banach-Saks target having bad weakly null trees with a weakly sequentially continuous coarse Lipschitz embedding whose increment tree realizes that bad behavior.

7. COUNTEREXAMPLE ROUTES CHECKED

This section records the most concrete routes examined, both to delimit the result and to make the remaining task reproducible.

The Kalton-Peck source candidate. The reflexive Kalton-Peck space Z_p fails the weak p -Banach-Saks property. Causey, Fovelle, and Lancien also show that Z_p fails the Hamming concentration properties indexed by p [5, Propositions 5.6–5.7]. Hence Z_p cannot coarse Lipschitz embed into targets carrying the corresponding tree/Hamming concentration. This does not exclude a weak p -Banach-Saks target with bad tree structure, so Z_p remains a natural possible source for a counterexample; no suitable weakly sequentially continuous embedding was found.

Braga's Baire-sum target candidate. The well-founded-tree construction in [4, Theorem 7.1] gives a separable reflexive target Y_p with weak p -Banach-Saks but with arbitrarily bad finite weakly null trees. It is therefore a natural target candidate. A counterexample would follow, for example, from a weakly sequentially continuous coarse Lipschitz embedding of ℓ_q into Y_p for some $1 < q < p$, since the unit vector basis shows that ℓ_q fails weak p -Banach-Saks. Natural support-sorting and branch-selection maps are not stable under changes in early coordinates and no coarse Lipschitz, weakly sequentially continuous construction was obtained.

Averaging and cube identities. The increment tree is a discrete gradient: endpoint values are independent of the order in which coordinates are added, and increments satisfy square-cocycle identities. Averaging over permutations or over finite blocks makes certain convex combinations of endpoint images small, using weak p -Banach-Saks locally. Coarse lower estimates, however,

apply to individual endpoints and do not pass from a small convex average to a small member without additional geometry of Y . This route therefore does not close the gap under the present hypothesis.

These failures are not counterexamples to the desired statement. They identify the two plausible fronts for further work: prove a branch estimate for gradient trees in every weak p -Banach-Saks space, or construct a WSC coarse Lipschitz map into a bad-tree weak p target.

8. LITERATURE STATUS AS OF JUNE 21, 2026

The source article appeared in the *Journal of the Institute of Mathematics of Jussieu* in 2021 and states Question 1.1 explicitly [4]. Searches were made for the exact problem number, the exact weak- p phrase, combinations with “weakly sequentially continuous” and “coarse Lipschitz”, and forward citations of the source. The later papers most closely connected to the problem develop tree- p -Banach-Saks and Hamming concentration properties [6], study the strict hierarchy of asymptotic properties and note that Z_p fails weak p -Banach-Saks [5], and construct concentration examples beyond asymptotic smoothness [7]. A 2025 monograph surveys the current nonlinear/asymptotic geometry program broadly [1].

No located paper states a proof or a counterexample to Question 1.1, and no located later source claims that Problem 8.6 has been resolved. On that evidence, the problem appears open. This is necessarily a literature-search conclusion rather than a proof of nonexistence: an unpublished or differently phrased result could have been missed.

9. CONCLUSION

The supplied partial result is correct and can be sharpened to the exact modulus estimate (2). In particular, tree- p -Banach-Saks is stable under weakly sequentially continuous coarse Lipschitz embeddings, with constants governed by the asymptotic slope ratio of the embedding. The original weak- p question is also positive for maps with controlled finite-sum defect.

The unresolved core is now quite narrow. Weak p -Banach-Saks must somehow control the special gradient trees generated by a single weakly sequentially continuous coarse Lipschitz map, even though it cannot control arbitrary weakly null trees. Alternatively, those gradient relations must be realized inside one of the known weak- p , bad-tree spaces to produce a counterexample.

REFERENCES

- [1] F. P. Baudier and G. Lancien, *Asymptotic and nonlinear geometries of Banach spaces and their interactions*, arXiv:2512.00817, 2025.
- [2] B. Beauzamy, Banach-Saks properties and spreading models, *Math. Scand.* **44** (1979), no. 2, 357–384.
- [3] B. de Mendonça Braga, Nonlinear weakly sequentially continuous embeddings between Banach spaces, *Int. Math. Res. Not. IMRN* (2020), no. 18, 5506–5533.
- [4] B. de Mendonça Braga, On asymptotic uniform smoothness and nonlinear geometry of Banach spaces, *J. Inst. Math. Jussieu* **20** (2021), no. 1, 65–102; arXiv:1808.03254.
- [5] R. M. Causey, A. Fovelle, and G. Lancien, Asymptotic smoothness in Banach spaces, three-space properties and coarse Lipschitz geometry, arXiv:2110.06710, version 2, 2022.
- [6] A. Fovelle, Hamming graphs and concentration properties in Banach spaces, arXiv:2106.04297, version 3, 2023.
- [7] A. Fovelle, Asymptotic smoothness, concentration properties in Banach spaces and applications, arXiv:2311.15900, version 2, 2024.